Nitrogen generation unit: operations overview

NitroChem Process Industries Limited (fictional). Synthetic data, Oct 2024 to Sep 2026.

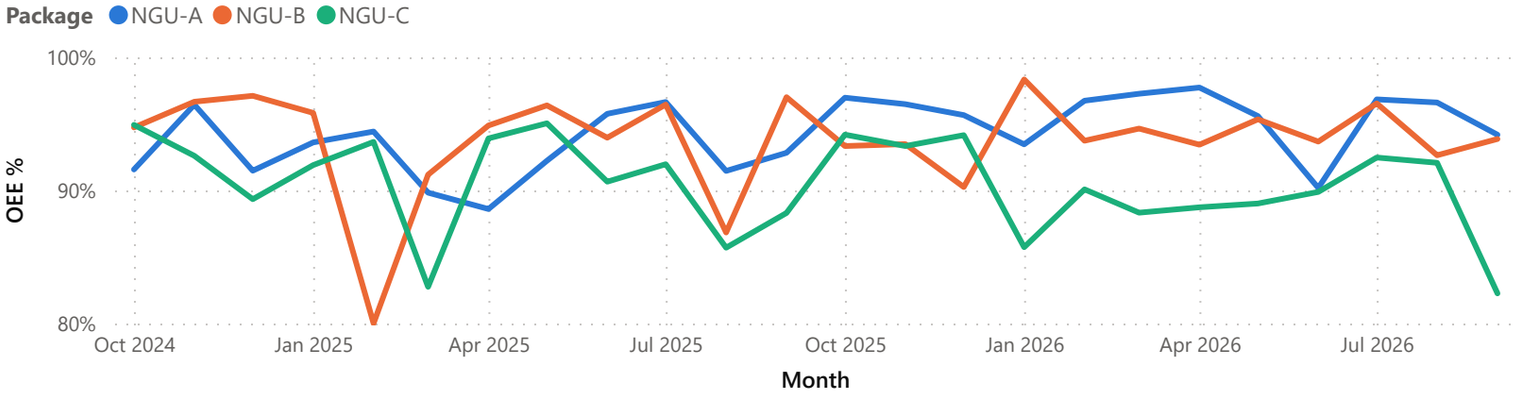
Month

All

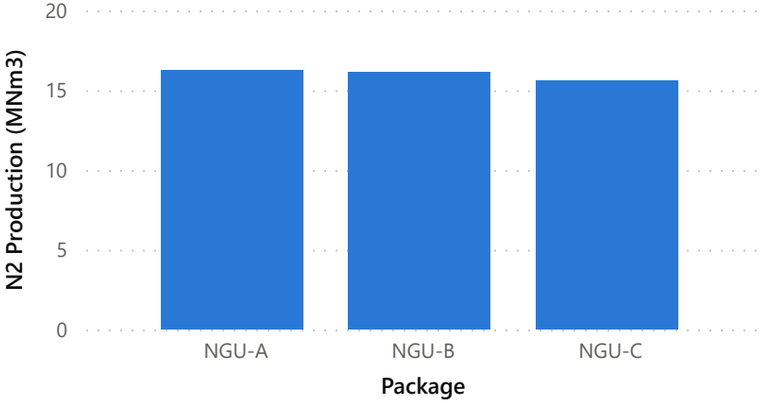
NGU-A | NGU-B | NGU-C

47.93	92.8%	1774	84	0.330	52
N2 Production (MNm3)	OEE %	Unplanned Downtime (h)	Trips	Energy Intensity (kWh/Nm3)	N2 Shortfall (h)

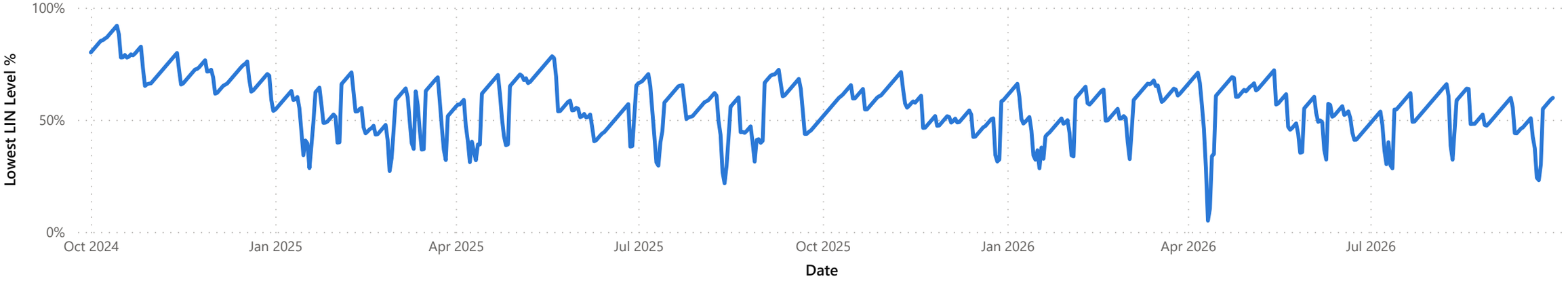
Monthly OEE by package



N2 production by package, million Nm3



Lowest LIN tank level each day



Downtime analysis

Unplanned and planned stops, two years. Gradual failures are the ones the anomaly model can catch.

Month

All

NGU-A | NGU-B | NGU-C

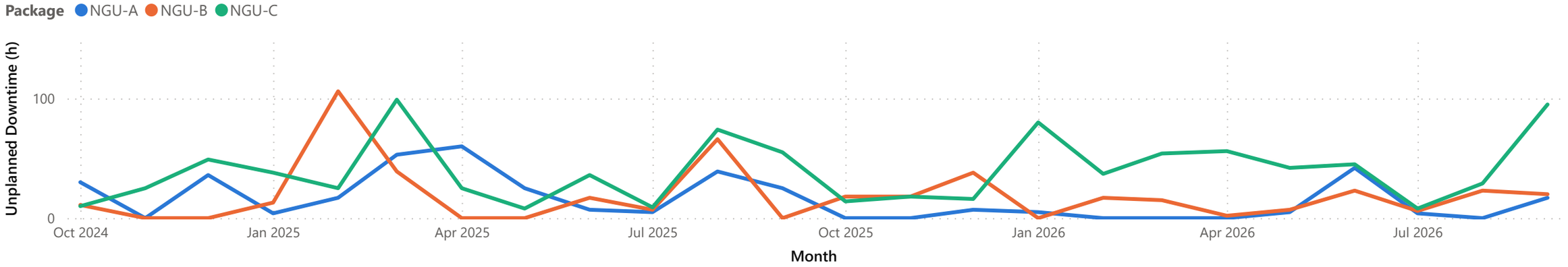
Failure onset

Gradual | Planned | Sudden

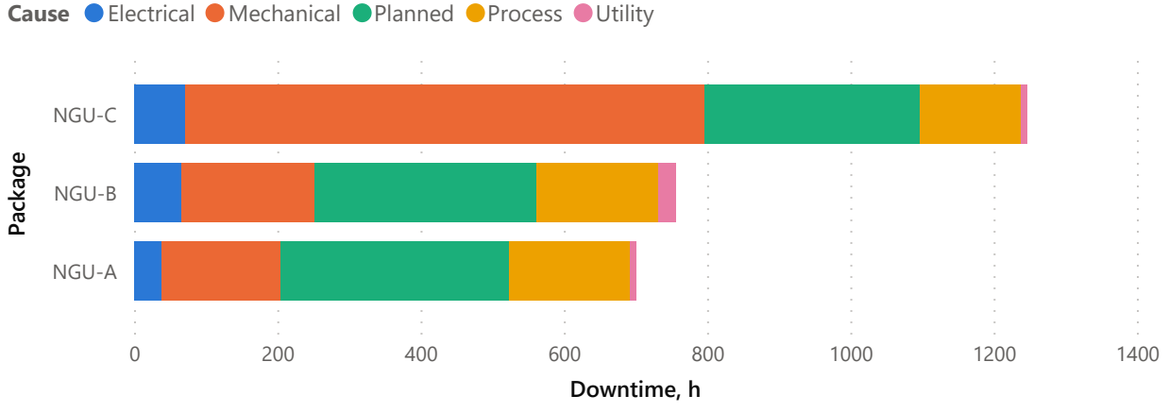
594
MTBF (h)

21.12
MTTR, h

Monthly unplanned downtime by package, hours



Downtime hours by package and cause



Unplanned downtime by failure mode

Failure mode	Trips	Downtime, h	MTTR, h
Compressor vibration	15	576	38.40
PPU breakthrough	18	455	25.28
Expander bearing	12	296	24.67
Coupling failure	4	204	51.00
Grid power dip	18	138	7.67
Cooling water loss	6	41	6.83
Motor relay trip	8	39	4.88
PPU valve failure	3	25	8.33
Total	**84**	**1774**	**21.12**

Energy efficiency

Compressor energy per Nm3 of on-spec N2. Power cost is an assumption; move the slider to change it.

Month

All

NGU-A | NGU-B | NGU-C

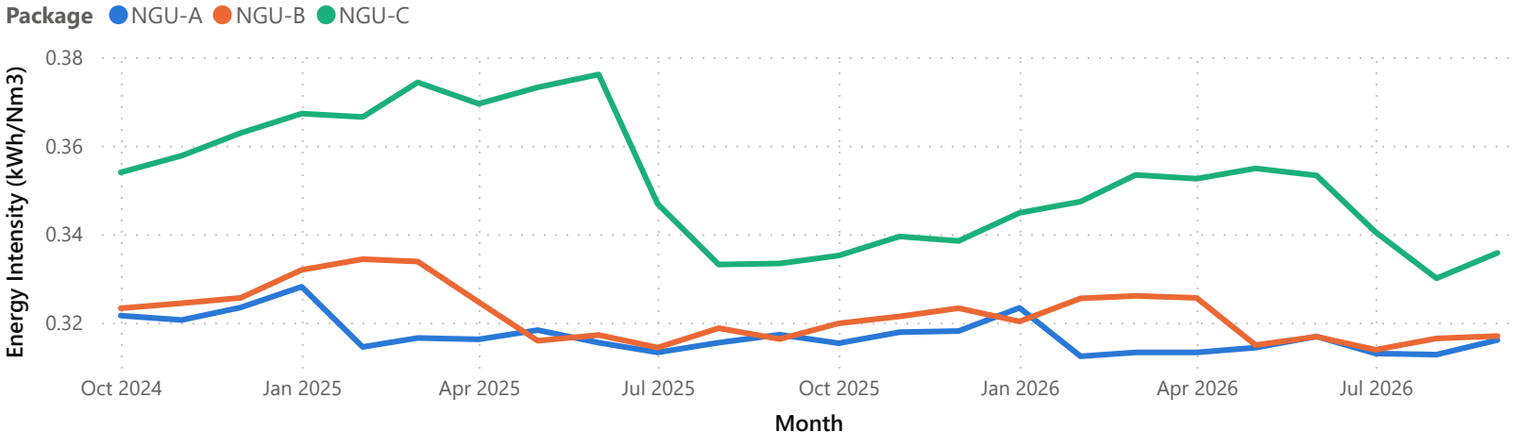
93.04

Excess Energy Cost (NGN m)

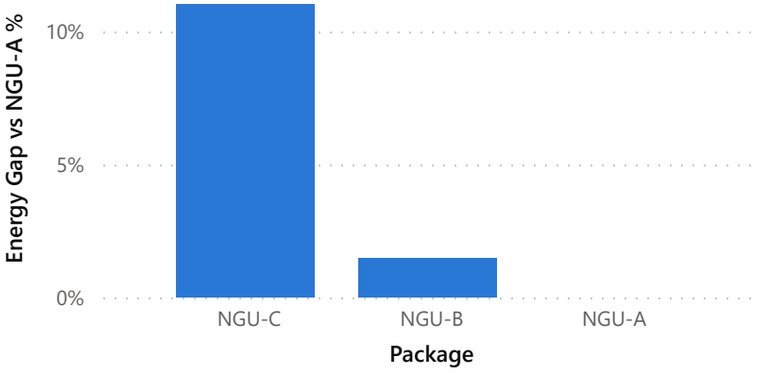
150

NGU-C's gap is about two thirds compressor wear and one third intercooler fouling (notebook 03).

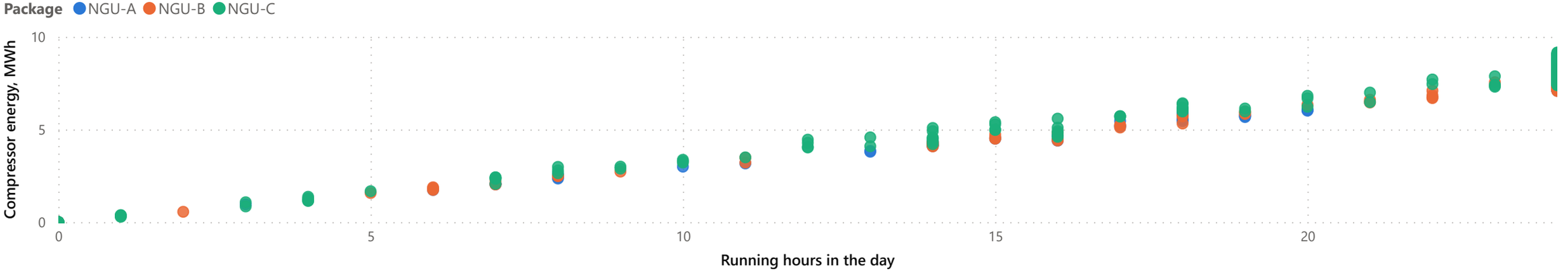
Monthly energy intensity by package, kWh per Nm3



Energy intensity gap vs NGU-A



Daily running hours vs compressor energy (one dot per package per day)



Anomaly alerts

Test period only, Feb to Sep 2026: months the model never saw in training.

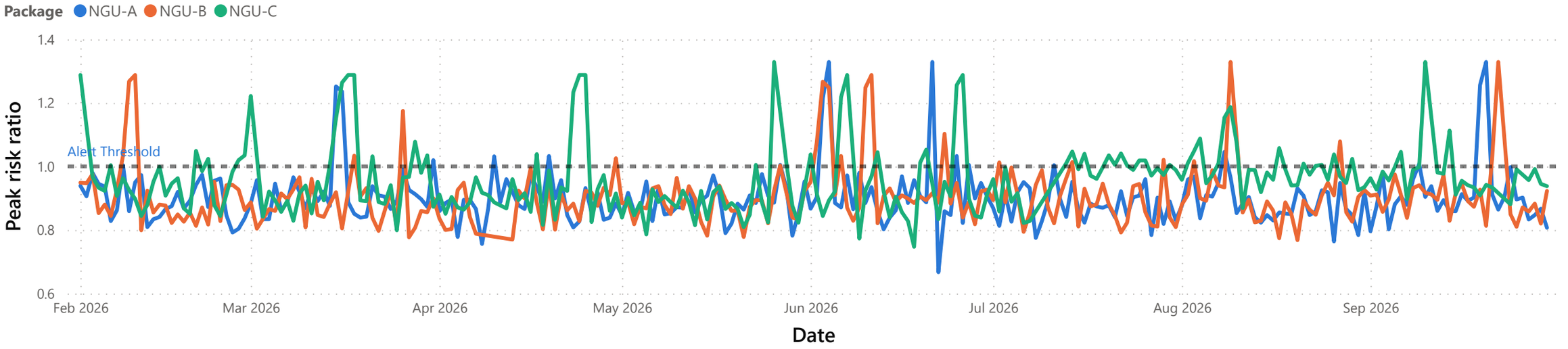
Month

All

NGU-A | NGU-B | NGU-C

Catches faults that develop over hours (bearings, compressor vibration, PPU breakthrough, cooling water). Cannot catch sudden failures: power dips, relay trips, coupling and valve failures.

Daily peak risk ratio by package (above 1.0 is anomalous)



High-risk days

Date	Package	Peak risk ratio	System flagged	Risk level	Alert hours
01 February 2026	NGU-C	1.29	Expander	High	7
09 February 2026	NGU-B	1.27	Expander	High	5
10 February 2026	NGU-B	1.29	Expander	High	5
28 February 2026	NGU-C	1.04	Compressor	High	2
01 March 2026	NGU-C	1.22	Compressor	High	18
15 March 2026	NGU-A	1.25	Expander	High	17
15 March 2026	NGU-C	1.15	Expander	High	8
16 March 2026	NGU-A	1.24	Expander	High	9
16 March 2026	NGU-C	1.26	Expander	High	24
17 March 2026	NGU-C	1.29	Expander	High	24
18 March 2026	NGU-C	1.29	Expander	High	12
26 March 2026	NGU-B	1.18	Compressor	High	5

7
False Alarms

17
Alerts Before Trip

46
High Risk Days